

ST117 Individual DRAFT Written Report - Phase 2 Task 1 (Woodland Vegetation – Tree Measurements)

Report Pod MS-05

2026-05-10

- Please replace XY-ab in filename.
- Please replace XY-ab, and 1999-12-31 in the header.
- Please read the document “WR Assessment, Style, and Submission Guidance” (see Moodle).
- By submitting this assignment, you confirm that you have read and will adhere to the guidance on AI set out in that document.
- Remember that you are responsible for everything that you submit.
- Please adhere to Word, page, figure, and table limits.
- Note that 20% of the total credit is dedicated to style.
- Please include R code in both Rmd and pdf files.

Contributions

This submission was created by (extend as needed): 1. WARWICK ID: u5683200

a)

include an output of three rows

```
wl_wide <- wl %>%
  filter(TREE_SPEC != 28 | is.na(TREE_SPEC)) %>%
  filter(FIELDNAME %in% c("DIAMETER", "HEIGHT", "DISTANCE", "NUM_STEMS")) %>%
  pivot_wider(
    id_cols = c(SITECODE, SYEAR, TREEID, STEMID, TREE_SPEC),
    names_from = FIELDNAME,
    values_from = VALUE,
    values_fn = mean
  ) %>%
  select(SITECODE, SYEAR, TREE_SPEC, DIAMETER, HEIGHT, DISTANCE, NUM_STEMS)
head(wl_wide, 3)
```

```
## # A tibble: 3 x 7
##   SITECODE SYEAR TREE_SPEC DIAMETER HEIGHT DISTANCE NUM_STEMS
##   <chr>    <int>    <int>    <dbl>  <dbl>    <dbl>    <dbl>
## 1 T05      1994      237      19     25      NA         1
## 2 T05      1994      237      16     22      NA         1
## 3 T05      1994      237      23     25      NA         1
```

b)

Limit: 50 words + 1 table.

```

model1 <- lm(HEIGHT ~ DIAMETER, data = wl_wide)
summary(model1)

##
## Call:
## lm(formula = HEIGHT ~ DIAMETER, data = wl_wide)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -52.531  -3.571   0.417   4.350  31.748
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  8.396693   0.301328   27.87  <2e-16 ***
## DIAMETER     0.108840   0.009056   12.02  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.672 on 801 degrees of freedom
## (1000 observations deleted due to missingness)
## Multiple R-squared:  0.1528, Adjusted R-squared:  0.1517
## F-statistic: 144.4 on 1 and 801 DF, p-value: < 2.2e-16

kable(summary(model1)$coefficients,
      digits = 3,
      caption = "Linear regression: Height ~ Diameter")

```

Table 1: Linear regression: Height ~ Diameter

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	8.397	0.301	27.866	0
DIAMETER	0.109	0.009	12.018	0

We select diameter as the predictor and height as the response variable. This is reasonable since taller trees tend to have wider trunks. The model is highly statistically significant ($p < 2.2e-16$), with a slope of 0.109 indicating an increase in height per unit diameter. However, the low R-squared (0.15) suggests that other factors such as species and site conditions also affect tree height.

c)

Limit: 400 words + 10 figures + 5 tables

```

model_T05 <- lm(HEIGHT ~ DIAMETER, data = wl_wide %>% filter(SITECODE == "T05"))
model_T06 <- lm(HEIGHT ~ DIAMETER, data = wl_wide %>% filter(SITECODE == "T06"))
model_T08 <- lm(HEIGHT ~ DIAMETER, data = wl_wide %>% filter(SITECODE == "T08"))
model_T09 <- lm(HEIGHT ~ DIAMETER, data = wl_wide %>% filter(SITECODE == "T09"))

```

```

ggplot(wl_wide, aes(x = DIAMETER, y = HEIGHT)) +
  geom_point(alpha = 0.3, colour = "steelblue") +
  geom_smooth(method = "lm", colour = "red") +
  labs(title = "Overall: Height ~ Diameter", x = "Diameter", y = "Height") +
  theme_minimal()

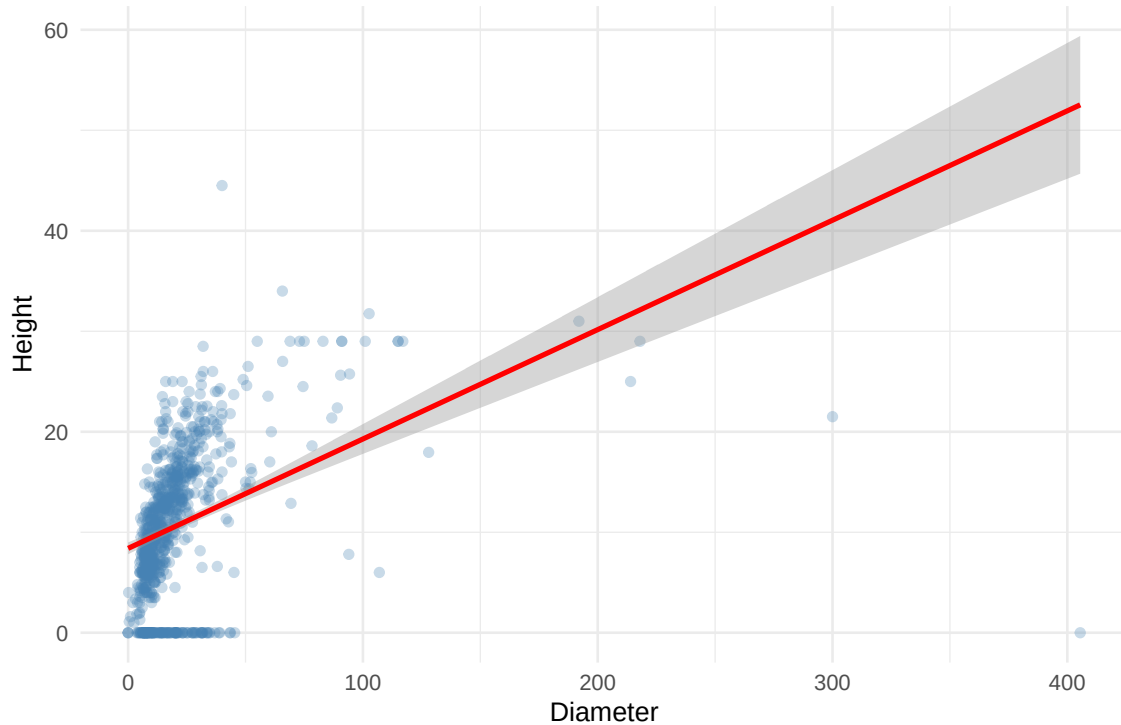
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## Warning: Removed 1000 rows containing non-finite outside the scale range
## (`stat_smooth()`).
```

```
## Warning: Removed 1000 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

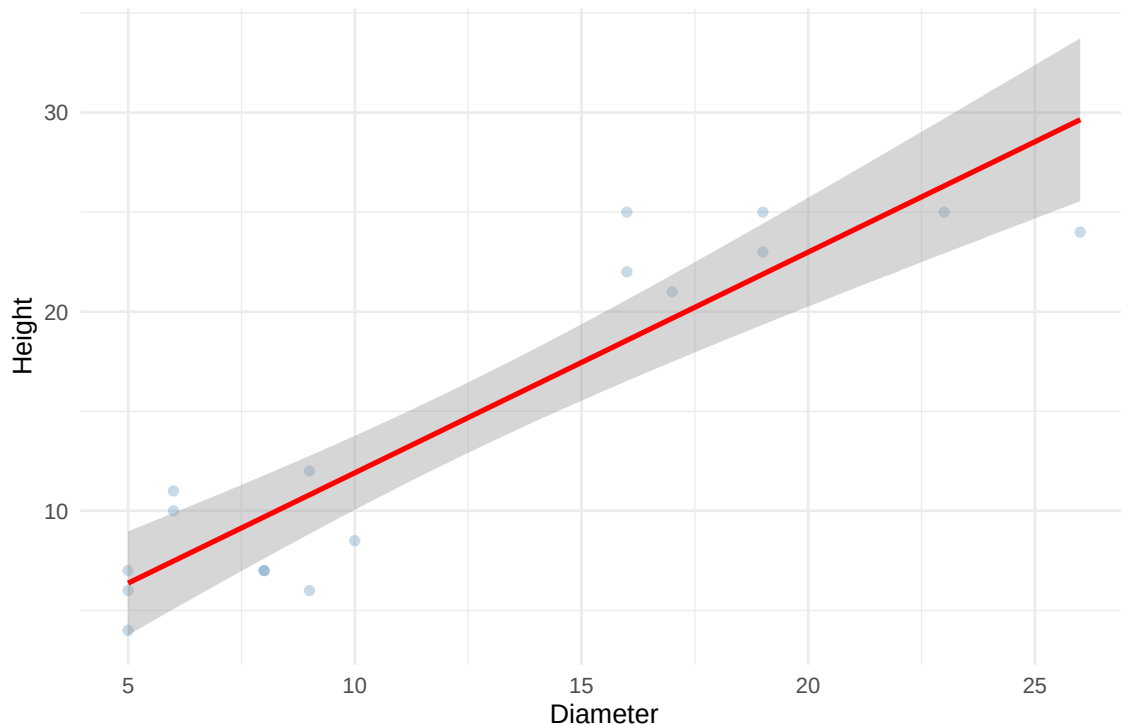
Overall: Height ~ Diameter



```
ggplot(wl_wide %>% filter(SITECODE == "T05"), aes(x = DIAMETER, y = HEIGHT)) +  
  geom_point(alpha = 0.3, colour = "steelblue") +  
  geom_smooth(method = "lm", colour = "red") +  
  labs(title = "T05: Height ~ Diameter", x = "Diameter", y = "Height") +  
  theme_minimal()
```

`geom\_smooth()` using formula = 'y ~ x'

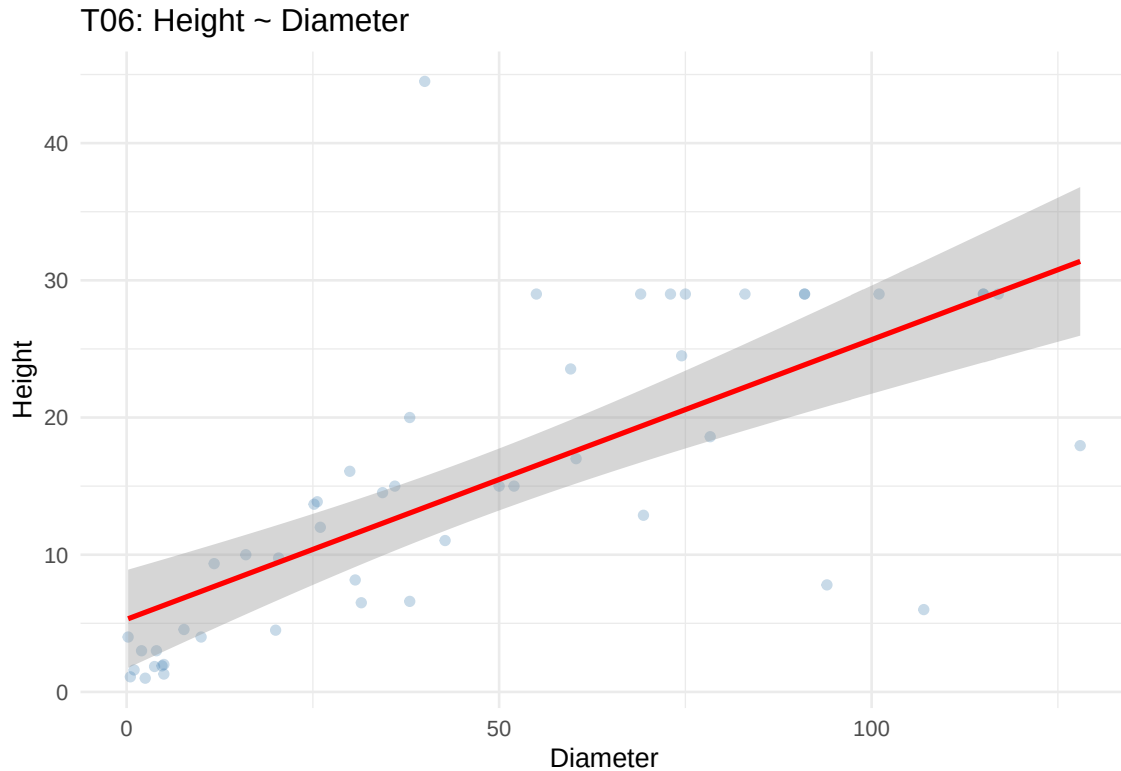
T05: Height ~ Diameter



```
ggplot(wl_wide %>% filter(SITECODE == "T06"), aes(x = DIAMETER, y = HEIGHT)) +  
  geom_point(alpha = 0.3, colour = "steelblue") +  
  geom_smooth(method = "lm", colour = "red") +
```

```
labs(title = "T06: Height ~ Diameter", x = "Diameter", y = "Height") +  
theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



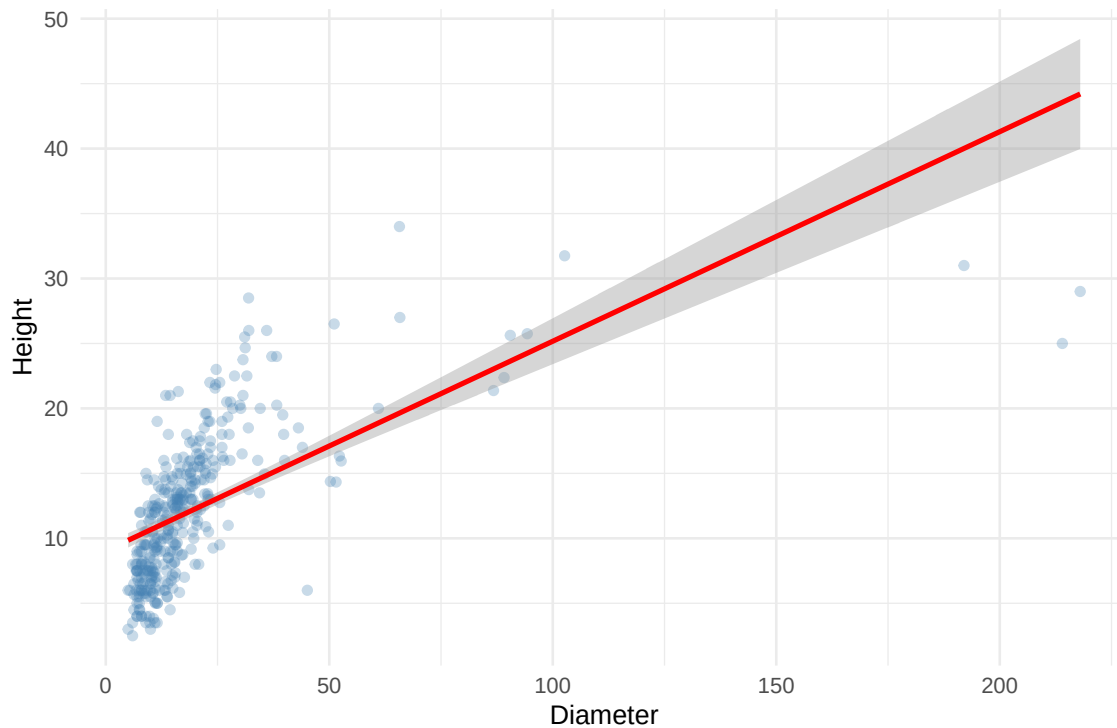
```
ggplot(wl_wide %>% filter(SITECODE == "T08"), aes(x = DIAMETER, y = HEIGHT)) +  
  geom_point(alpha = 0.3, colour = "steelblue") +  
  geom_smooth(method = "lm", colour = "red") +  
  labs(title = "T08: Height ~ Diameter", x = "Diameter", y = "Height") +  
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## Warning: Removed 698 rows containing non-finite outside the scale range  
## (`stat_smooth()`).
```

```
## Warning: Removed 698 rows containing missing values or values outside the scale range  
## (`geom_point()`).
```

T08: Height ~ Diameter



```
ggplot(wl_wide %>% filter(SITECODE == "T09"), aes(x = DIAMETER, y = HEIGHT)) +  
  geom_point(alpha = 0.3, colour = "steelblue") +  
  geom_smooth(method = "lm", colour = "red") +  
  labs(title = "T09: Height ~ Diameter", x = "Diameter", y = "Height") +  
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

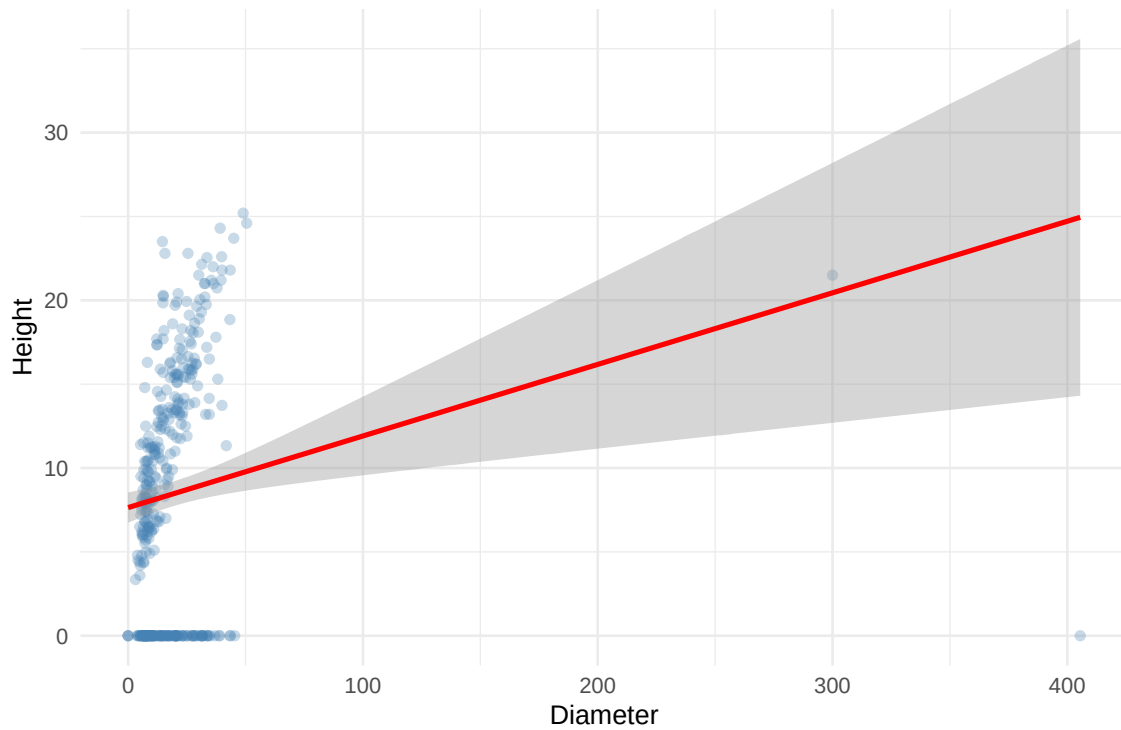
```
## Warning: Removed 302 rows containing non-finite outside the scale range
```

```
## (`stat_smooth()`).
```

```
## Warning: Removed 302 rows containing missing values or values outside the scale range
```

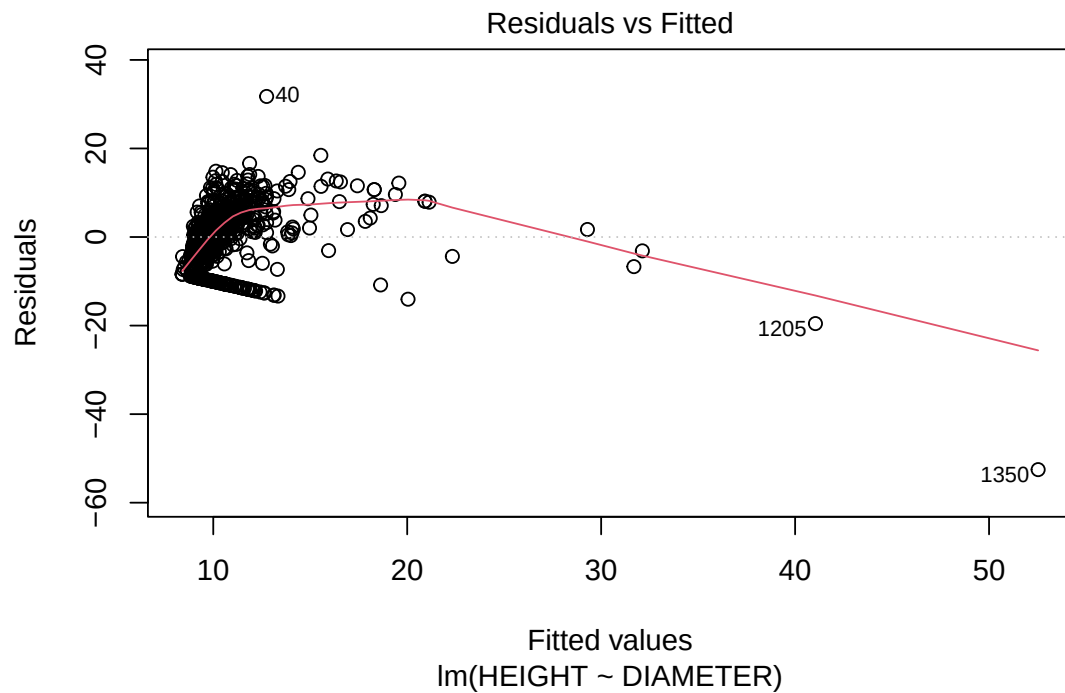
```
## (`geom_point()`).
```

T09: Height ~ Diameter



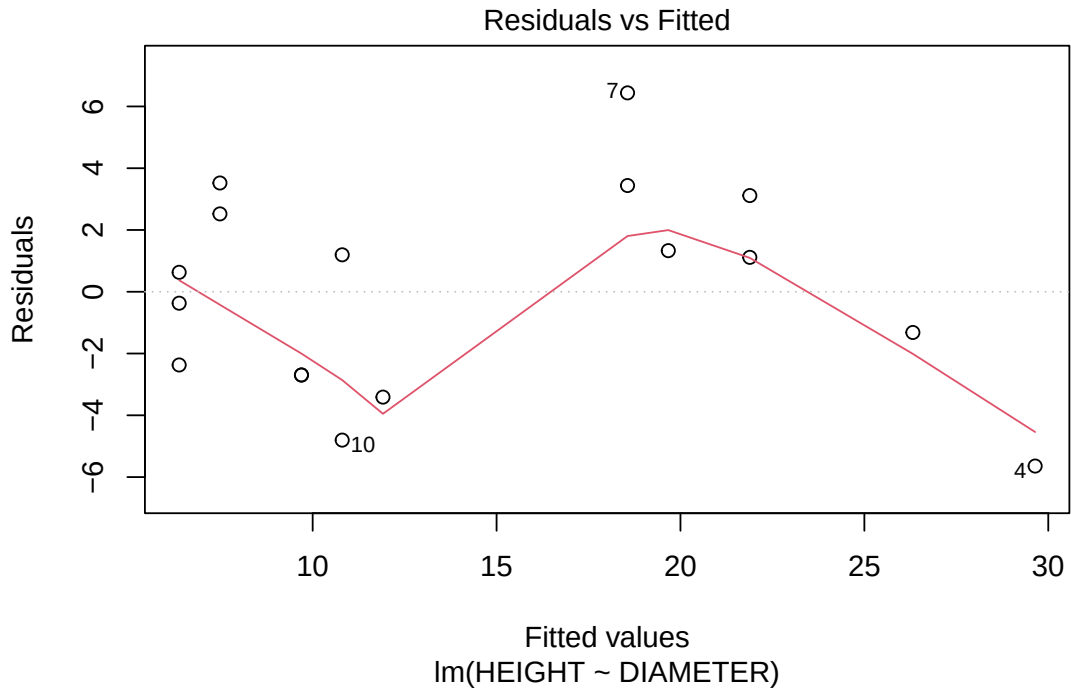
```
plot(model1, which = 1, main = "Overall: Residuals vs Fitted")
```

Overall: Residuals vs Fitted



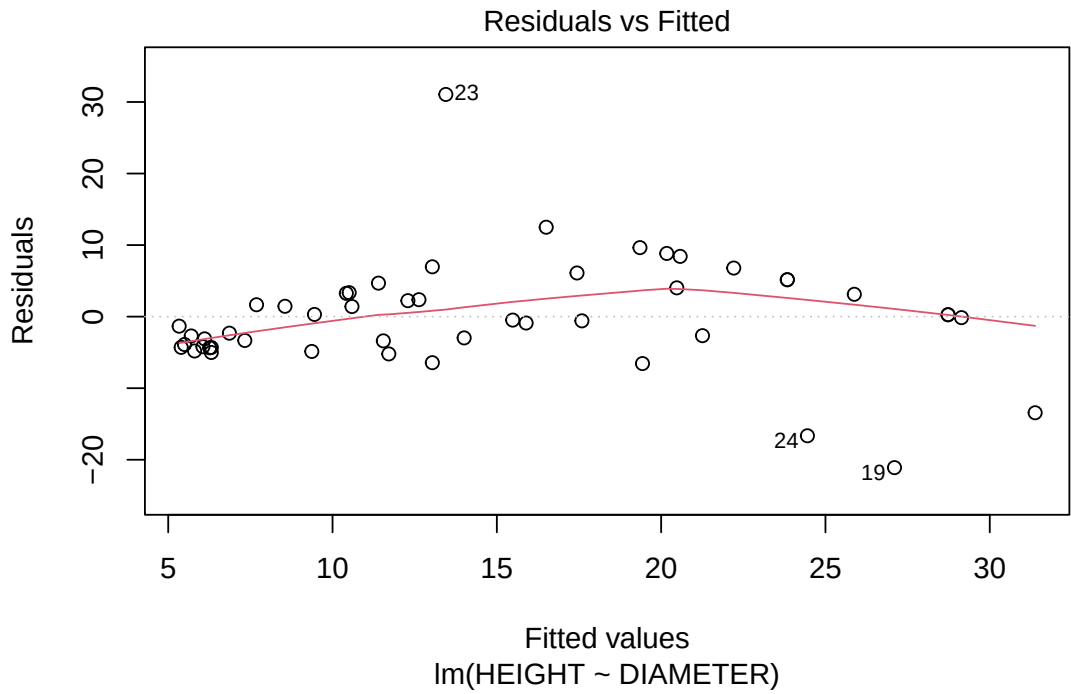
```
plot(model_T05, which = 1, main = "T05: Residuals vs Fitted")
```

T05: Residuals vs Fitted



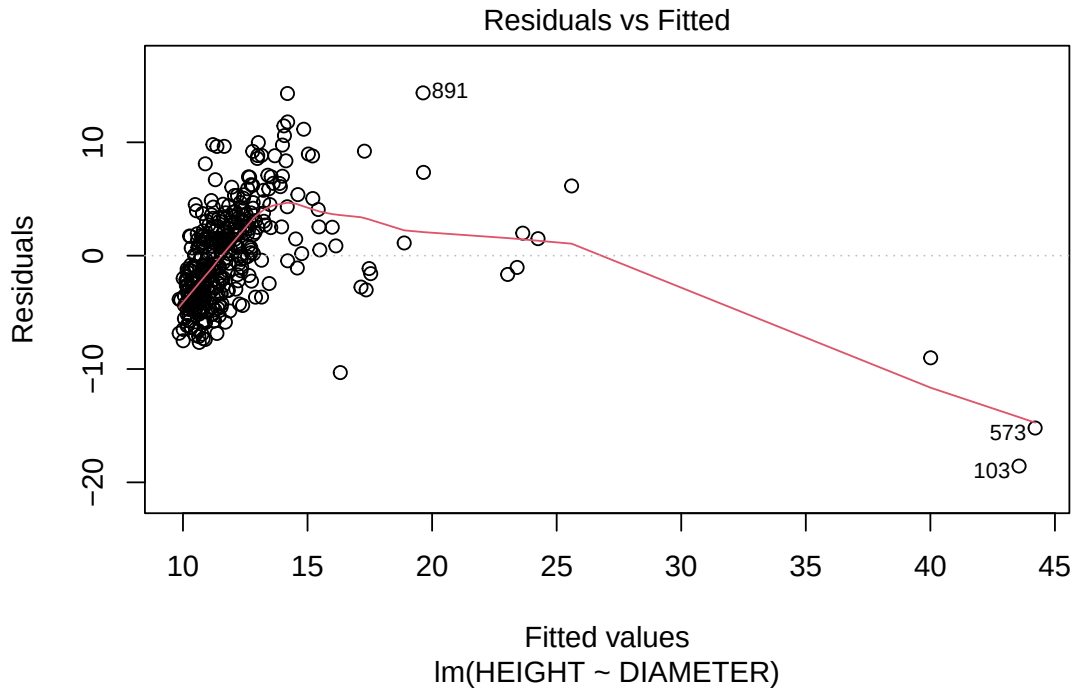
```
plot(model_T06, which = 1, main = "T06: Residuals vs Fitted")
```

T06: Residuals vs Fitted



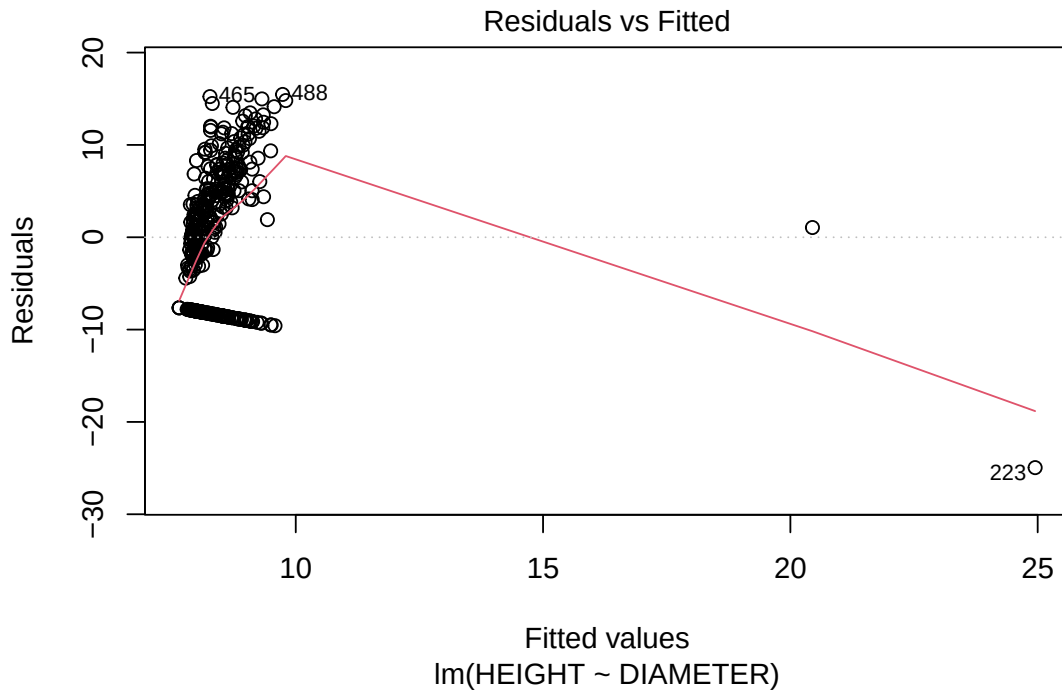
```
plot(model_T08, which = 1, main = "T08: Residuals vs Fitted")
```

T08: Residuals vs Fitted



```
plot(model_T09, which = 1, main = "T09: Residuals vs Fitted")
```

T09: Residuals vs Fitted



```
kable(summary(model_T05)$coefficients, digits = 3, caption = "T05: Height ~ Diameter")
```

Table 2: T05: Height ~ Diameter

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.829	1.736	0.478	0.64
DIAMETER	1.108	0.125	8.848	0.00


```
kable(summary(model_T06)$coefficients, digits = 3, caption = "T06: Height ~ Diameter")
```

Table 3: T06: Height ~ Diameter

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.292	1.783	2.967	0.005
DIAMETER	0.204	0.030	6.794	0.000

```
kable(summary(model_T08)$coefficients, digits = 3, caption = "T08: Height ~ Diameter")
```

Table 4: T08: Height ~ Diameter

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	9.040	0.315	28.653	0
DIAMETER	0.161	0.011	14.944	0

```
kable(summary(model_T09)$coefficients, digits = 3, caption = "T09: Height ~ Diameter")
```

Table 5: T09: Height ~ Diameter

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	7.643	0.458	16.693	0.000
DIAMETER	0.043	0.014	3.058	0.002

```
data.frame(
  Model = c("Overall", "T05", "T06", "T08", "T09"),
  R_squared = c(
    summary(model1)$r.squared,
    summary(model_T05)$r.squared,
    summary(model_T06)$r.squared,
    summary(model_T08)$r.squared,
    summary(model_T09)$r.squared
  ),
  Slope = c(
    coef(model1)[2],
    coef(model_T05)[2],
    coef(model_T06)[2],
    coef(model_T08)[2],
    coef(model_T09)[2]
  )
) %>% kable(digits = 3, caption = "Model comparison across sites")
```

Table 6: Model comparison across sites

Model	R_squared	Slope
Overall	0.153	0.109
T05	0.839	1.108
T06	0.495	0.204
T08	0.382	0.161
T09	0.025	0.043

All five models show a positive relationship between diameter and height, consistent with expected tree growth where larger diameter trees tend to be taller. However, model fit varies quite a lot across sites, suggesting that the relationship between diameter and height is not uniform across the woodland locations monitored by the ECN. T05 shows the strongest fit ($R^2 = 0.84$), with

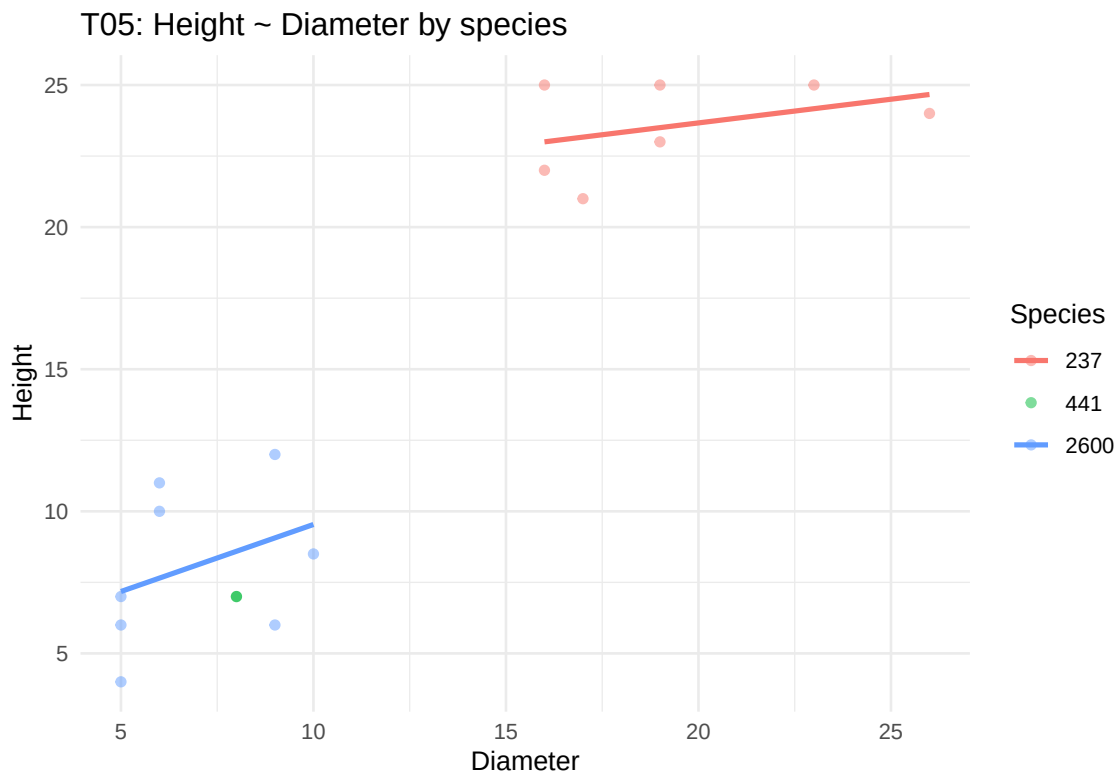
a slope of 1.108, suggesting that diameter is a very strong predictor of height at this site. T06 shows moderate fit ($R^2 = 0.50$) with a shallower slope of 0.204, while T08 is weaker still ($R^2 = 0.38$, slope = 0.161), suggesting that at these sites other factors beyond diameter contribute more to variation in height. T09 shows almost no relationship between diameter and height ($R^2 = 0.025$, slope = 0.043), suggesting that diameter is a very poor predictor of height at this site. This could reflect a more diverse tree species composition at T09, where different species have different growth forms, or it could reflect differences environmental conditions that affect diameter and height growth. The overall model ($R^2 = 0.15$) performs poorly compared to the site-level models, demonstrating that pooling data across sites masks important site-level variation in the diameter-height relationship. The residual plots for T08 and T09 closely resemble the overall model, reflecting their larger contribution to the pooled dataset, while T05 and T06 show distinct residual patterns.

d)

Limit: 300 words + 4 figures + 4 tables

```
wl_wide %>%
  filter(SITECODE == "T05") %>%
  ggplot(aes(x = DIAMETER, y = HEIGHT, colour = factor(TREE_SPEC))) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "T05: Height ~ Diameter by species",
       x = "Diameter", y = "Height", colour = "Species") +
  theme_minimal()
```

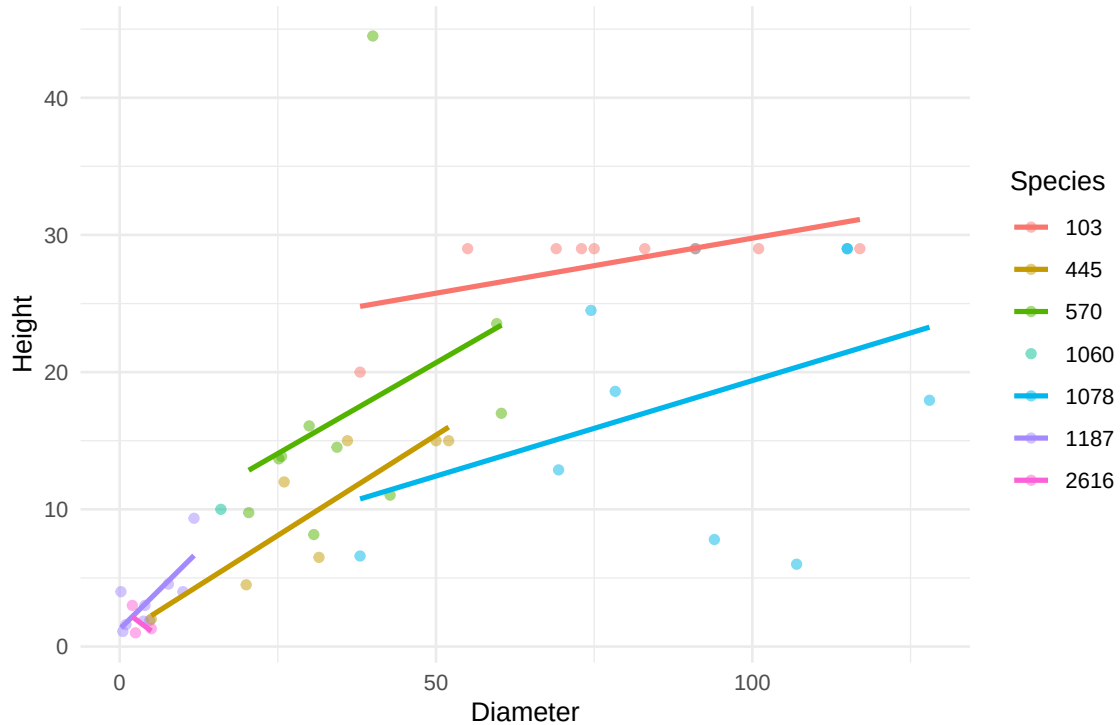
`geom\_smooth()` using formula = 'y ~ x'



```
wl_wide %>%
  filter(SITECODE == "T06") %>%
  ggplot(aes(x = DIAMETER, y = HEIGHT, colour = factor(TREE_SPEC))) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "T06: Height ~ Diameter by species",
       x = "Diameter", y = "Height", colour = "Species") +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

T06: Height ~ Diameter by species



```
wl_wide %>%
  filter(SITECODE == "T08") %>%
  ggplot(aes(x = DIAMETER, y = HEIGHT, colour = factor(TREE_SPEC))) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "T08: Height ~ Diameter by species",
       x = "Diameter", y = "Height", colour = "Species") +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

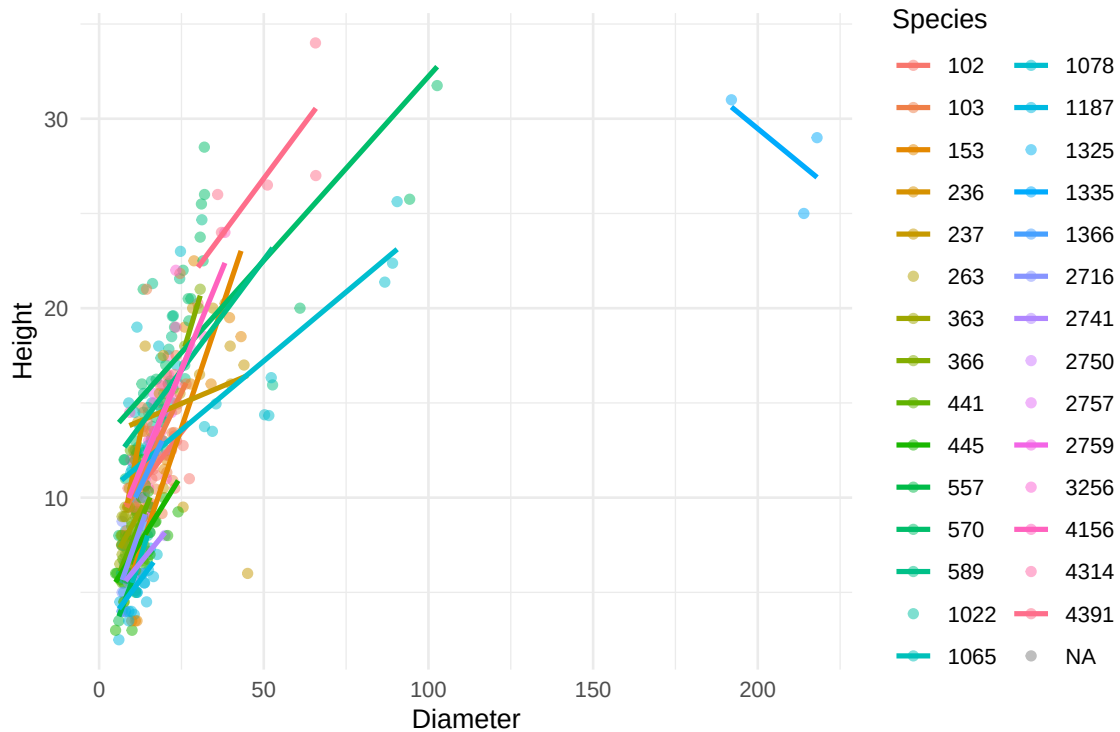
```
## Warning: Removed 698 rows containing non-finite outside the scale range
```

```
## (`stat_smooth()`).
```

```
## Warning: Removed 698 rows containing missing values or values outside the scale range
```

```
## (`geom_point()`).
```

T08: Height ~ Diameter by species



```
wl_wide %>%
  filter(SITECODE == "T09") %>%
  ggplot(aes(x = DIAMETER, y = HEIGHT, colour = factor(TREE_SPEC))) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "T09: Height ~ Diameter by species",
       x = "Diameter", y = "Height", colour = "Species") +
  theme_minimal()
```

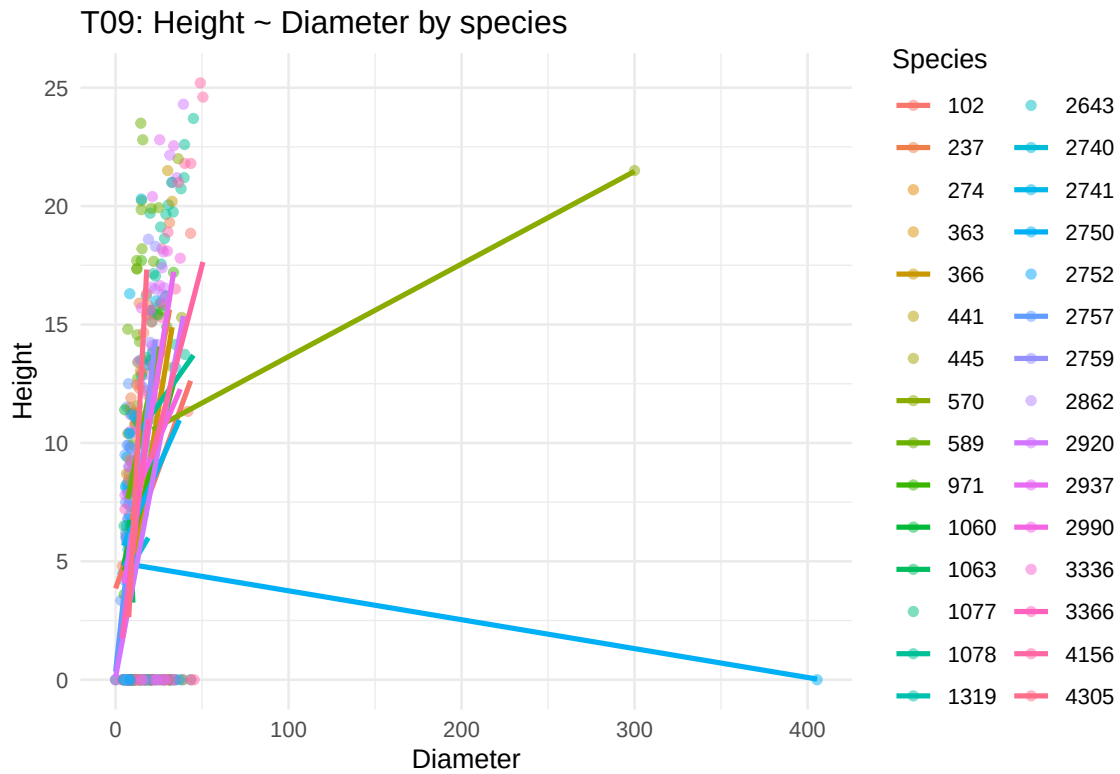
```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## Warning: Removed 302 rows containing non-finite outside the scale range
```

```
## (`stat_smooth()`).
```

```
## Warning: Removed 302 rows containing missing values or values outside the scale range
```

```
## (`geom_point()`).
```



```
model_T05_sp <- lm(HEIGHT ~ DIAMETER + factor(TREE_SPEC),
  data = wl_wide %>% filter(SITECODE == "T05"))
model_T06_sp <- lm(HEIGHT ~ DIAMETER + factor(TREE_SPEC),
  data = wl_wide %>% filter(SITECODE == "T06"))
model_T08_sp <- lm(HEIGHT ~ DIAMETER + factor(TREE_SPEC),
  data = wl_wide %>% filter(SITECODE == "T08"))
model_T09_sp <- lm(HEIGHT ~ DIAMETER + factor(TREE_SPEC),
  data = wl_wide %>% filter(SITECODE == "T09"))

data.frame(
  Site = c("T05", "T06", "T08", "T09"),
  R2_without_species = c(
    summary(model_T05)$r.squared,
    summary(model_T06)$r.squared,
    summary(model_T08)$r.squared,
    summary(model_T09)$r.squared),
  R2_with_species = c(
    summary(model_T05_sp)$r.squared,
    summary(model_T06_sp)$r.squared,
    summary(model_T08_sp)$r.squared,
    summary(model_T09_sp)$r.squared)
) %>% kable(digits = 3, caption = "R-squared before and after including tree species")
```

Table 7: R-squared before and after including tree species

Site	R2_without_species	R2_with_species
T05	0.839	0.942
T06	0.495	0.691
T08	0.382	0.770
T09	0.025	0.130

A key feature affecting simple linear regression at all four sites is the presence of multiple tree species with distinct diameter-height relationships. Different species have different growth forms, meaning a single regression line fitted across all species will not be able to capture the true relationships within each species. At T05, species 2600 shows a much steeper diameter-height slope than species

237, while species 441 has only a single observation, making species-level analysis unreliable for rare species. At T06, the dominant species 570 drives the overall relationship while other species show different patterns. T08 is the most species-diverse site with 30+ species, explaining its moderate R-squared of 0.38. The many different growth forms create considerable scatter around a single regression line. At T09, despite species 292 dominating, the near-zero R-squared (0.025) suggests that even within sites, species differences overwhelm any simple diameter-height relationship. To account for this, we refit the model including TREE_SPEC as an additional categorical predictor. This substantially improves model fit across all sites. R-squared increases from 0.84 to 0.94 at T05, 0.50 to 0.69 at T06, 0.38 to 0.77 at T08, and 0.025 to 0.13 at T09. While T09 remains weak even after accounting for species, the improvements at other sites confirm that species identity is an important predictor of the diameter-height relationship.